

No. of Functional Features Included in the Solution

Date	14 July, 2026
Team ID	
Project Name	Credit Card Approval Screening
Maximum Marks	5 Marks

Functional Features Overview

This document lists the functional features implemented in the Credit Card Approval Screening project, based on the delivered **app.py** Flask application. Each feature contributes to scoring applicants with a trained Machine Learning pipeline and surfacing that decision through a Flask web application for customers, analysts, and compliance officers.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status (Done / in progress / Pending)	Marks Contribution
1	Applicant Data Input	Accepts applicant details (income, employment, housing, etc.) through the / form, rendered with category and numeric field options.	Flask UI (index route)	Done	High
2	Data Preprocessing & Feature Encoding	build_feature_row() converts a raw form/JSON payload into a one-hot-encoded row aligned to the trained feature columns, then scales it with the saved scaler.	Data Processing	Done	High
3	Credit Risk Prediction	score_application() runs the scaled row through the trained model and returns an Approve/Decline decision with the approval and high-risk probabilities.	Prediction Engine (/predict)	Done	High
4	Model Explainability	explain() returns the top contributing features for a prediction using SHAP when available, falling back to the model's global feature importance otherwise.	Prediction Engine	Done	Medium
5	Borderline Case Detection & Analyst Queue	Predictions with a high-risk probability between 0.40 and 0.60 are flagged as borderline and appended to an in-memory queue for analyst review.	Backend / Workflow	Done	Medium
6	Model Comparison Dashboard	/dashboard renders model comparison and income-vs-approval charts, powered by JSON served from /api/metrics (computed from metrics.json and applicants.csv).	Frontend + API / Integration	Done	Medium
7	What-If Scenario Sandbox	/sandbox lets a customer test different applicant scenarios against the model without saving or persisting any data.	Frontend	Done	Medium
8	Model Artifact Loading & Integration	Trained model, scaler, feature columns, metrics, and (optionally) the SHAP background dataset are loaded once at startup via joblib and shared across all routes.	Backend	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Applicant Data Input, Data Preprocessing & Feature Encoding, Credit Risk Prediction, Model Artifact Loading & Integration
Additional / Nice-to-Have Features	Model Explainability, Borderline Case Detection & Analyst Queue, Model Comparison Dashboard, What-If Scenario Sandbox
Features Tested & Verified	8 features reviewed and confirmed working by code inspection; app.py does not include an automated test suite, so results are not backed by unit tests.

Feature Category Breakdown

S.No	Category	Features in Category	Example Feature
1	User Interface (UI)	2	Applicant Data Input Form, What-If Scenario Sandbox
2	Backend / Logic	3	Data Preprocessing & Feature Encoding, Borderline Case Detection & Analyst Queue, Model Artifact Loading & Integration
3	Prediction / ML Engine	2	Credit Risk Prediction, Model Explainability
4	API / Integration	1	Model Comparison Dashboard (/dashboard + /api/metrics)
5	Security / Authentication	0	Not implemented in app.py — no login, session, or role-based access control on analyst/compliance routes